# FAITH OGUNDIMU

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## PROFESSIONAL SUMMARY


Computational cancer-genomics PhD researcher and Government of Ireland Scholar, developing machine-learning and multi-omic methods to find non-coding drivers of cancer. First-author review in minor revisions at npj Genomic Medicine (Nature portfolio), with work spanning chromatin accessibility, cfDNA methylation, single-cell regulatory genomics and CRISPR-based validation across thousands of breast-cancer genomes. Turns large, noisy biological data into interpretable models, rigorous evaluation and testable candidates. Committed to science communication, patient and public involvement, and broadening access to advanced genomics.


## EDUCATION

**Royal College of Surgeons Ireland (RCSI)** Sep 2025 - Aug 2029

Doctor of Philosophy (PhD) in Cancer Genomics and Epigenomics, Machine Learning and Bioinformatics

Thesis: *Identification of non-coding somatic driver mutations in breast cancer*

**Dublin City University (DCU)** Sep 2021 - May 2025

BSc. in Genetics and Cell Biology (GCB) | First Class Honours (1.1) | Ranked 2nd overall (Salutatorian) & Dean's Honour List

Final Dissertation: *Using neural networks to understand gene regulation in the MCF-7 breast cancer cell line*

## RESEARCH EXPERIENCE

**Royal College of Surgeons Ireland (RCSI)** Sep 2025 - Aug 2029

**Doctoral Researcher**

- Building machine-learning models to pick out the non-coding mutations that drive breast cancer from more than 5,000 tumour genomes, then testing the top candidates with CRISPR in cell lines.
- Mapping, in 3D and tissue-by-tissue, the gene-control regions of the 98% of the genome that does not code for protein, where most cancer drivers are still unknown.

**Unaite Fellowship for AI in Biology | Paris, France (hybrid)** Sep 2026 - present

Fellow, Inaugural Cohort (1 of 20) - project mentored by Cécile Badoual, Quentin Garrido and Jérémie Kalfon

- Building an interpretable AI model that reconstructs a cell-by-cell map of the tumour microenvironment from incomplete spatial data, fusing routine pathology images, spatial gene-expression maps and single-cell protein measurements even when one modality is missing or only loosely matched.
- Largely Paris-based across the fellowship year on a hybrid basis, with regular in-person residencies, developing the representation-learning and uncertainty methods that keep the reconstruction reliable and validating it against matched single-cell protein data.

**Institut Curie (Bioinformatics, Biostatistics, Epidemiology & Computational Systems / CUBIC)** Sep 2026 - Oct 2026

Visiting Researcher - France Excellence Research Residency (host: Dr Nicolas Servant)

- Built the 3D-genome arm of my PhD: reprocessed and copy-number-corrected Hi-C contact maps for three breast-cancer subtypes, and showed that regulatory regions active in a given subtype physically fold toward their target genes more in that subtype, evidence that 3D structure, not just DNA sequence, carries subtype-specific gene control.
- Produced the project's first map of the structural boundaries that divide the genome into regulatory neighbourhoods, now a reference map that identifies which gene a regulatory element actually controls rather than assuming it is the nearest one.

**Beaumont RCSI Cancer Centre, Irish Cancer Society, Lung Health Check** Jan 2026 - Jul 2026

Bioinformatics Research Assistant – BRAND Study (NSCLC Liquid Biopsy & cfDNA Methylation Analysis)

- Built an end-to-end pipeline across 88 blood-plasma samples to separate responders from non-responders to immunotherapy in advanced lung cancer, narrowing millions of DNA-methylation sites to a 100-marker panel.
- Showed the circulating tumour signal falls after treatment and tracks patient survival, for one of Ireland's largest lung-cancer screening pilots (€4.9M).

**University College Dublin** Jun 2025 - Aug 2025

Breakthrough Summer Research Scholar (BioSLATE - Biomarker Selection and Synthetic Lethality Analysis for Therapeutic Exploration in high-grade serous ovarian cancer (HGSOC)) - BioSLATE Website | GitHub (BioSLATE) | GitHub (OncoSynth)

- Identified synthetic-lethal targets in HGSOC via significant CNA–CRISPR correlations across >500 tumours, integrating TCGA CNA, DepMap CRISPR and gene-essentiality models; shipped an interactive Streamlit app for target nomination.
- Built OncoSynth, a CrewAI multi-agent CLI automating literature mining, druggability scoring and clinical-trial linkage for translational target discovery.

**Royal College of Surgeons Ireland (Final Year Dissertation)** Jan 2025 - Mar 2025

Visiting Student Researcher - GitHub

- Trained a neural network to predict which regions of the breast-cancer genome are open and active from histone marks and transcription-factor signals; it beat a published model while using fewer inputs and no DNA sequence.
- Found that a single mark (H3K4me1) did most of the work, with histone marks alone nearly matching the full model.

## PUBLICATIONS

Ogundimu F, Toomey S & Furney SJ: Deciphering the Non-coding Cancer Genome through Deep Learning and Multiomic Integration (minor revision; revised manuscript resubmitted to npj Genomic Medicine, September 2026)

Ogundimu F†, Cenci C†, Huang Y & Giannarou S: A Second Look: Action-Conditioned Distance Regression for Robot-Assisted Endomicroscopy (submitted to IEEE ISBI 2027). †Equal contribution.

O'Reilly D, Ogundimu F, et al.: Multimodal genomic features in exhaled breath and plasma to predict outcomes to immune checkpoint inhibition in advanced non-small cell lung cancer *(submitted to Clinical Cancer Research)*

Thinnes C. et al.: DrugMap: A genome-scale visualisation of human drug metabolism *(in preparation)*

## ADDITIONAL RESEARCH & PROFESSIONAL EXPERIENCE

**Onkydra** May 2026 - Present

**Lead Developer (early-stage project)**

- Developing Onkydra, an in-silico target-validation tool for rare and paediatric cancers, built on public data from 60 H3 K27M diffuse midline glioma cases; it applies the non-coding regulatory methods of my research group and is being taken forward in consultation with RCSI's innovation office.
- Category finalist, Build with Gemini XPRIZE 2026 (one of 20 in Professional Services Access).

**Virtual Metabolic Human (Virtuome)** May 2024 - Jul 2024

Research Intern

- Built interactive visualisations of >1,000 drug-metabolism pathways for the Virtual Metabolic Human, making dense metabolic networks navigable for researchers and improving pipeline efficiency by ~45%.
- Delivered a community-impact project with the Coeliac Society of Ireland, using Public and Patient Involvement (PPI) to survey the coeliac community and translate the findings into a patient-facing report and infographic, informed by lived experience of coeliac disease.

**Applied Process Company (APC) Ltd.** Jan 2024 - Aug 2024

Analytical Intern

- Independently executed plasmid DNA extraction and GLP-compliant quality-control workflows for high-priority biopharma projects (~80% of assays autonomously), using HPLC and Agilent OpenLab ChemStation.

**Royal Victoria Eye and Ear Hospital** Jul 2023 - Sep 2023

Histopathology Summer Intern

- Streamlined histopathology workflows and digital tracking (Clinisys WinPath), cutting turnaround times ~60%; conducted microtomy and H&E staining and supported research into uveal melanoma prognostics in young patients in Ireland.

**PRODIGY Creative** 2020 - 2023

Founder & Creative Designer

- Founded and ran a multimedia design business, delivering 500+ projects for 300+ clients worldwide in both English and French, before transitioning fully into bioinformatics and cancer research.

## GRANTS & SCHOLARSHIPS

**Government of Ireland (GOI) Postgraduate Scholarship (€136,000, 18% success rate) –** Nationally competitive award recognising exceptional research potential.

**France Excellence Research Residency Grant (€1,850, French Embassy in Ireland) –** Competitively awarded to one of nine awardees nationally; funding a 1-month research residency at Institut Curie (CUBIC, Dr. Nicolas Servant) for CNV-aware chromatin topology analysis and AI-ready regulatory map construction in breast cancer.

**RCSI Top-Up Research Grant (€10,000) –** Selective supplement to GOI Scholarship supporting CRISPR-based validation of computationally predicted non-coding driver mutations in breast cancer.

**Breakthrough Cancer Research Summer Scholarship (€3,000) –** National cancer research award supporting 12 hands-on projects in poor-prognosis cancers.

**Unaite Fellowship for AI in Biology (Paris, 2026-27) –** Selected for the inaugural cohort of a competitive year-long fellowship for researchers at the intersection of AI and biology, mentored by leaders in foundation models for biology.

## SELECTED OPEN-SOURCE TOOLS & PROJECTS

**NCypher - 1st place, Researcher track, Built with Claude: Life Sciences** Jul 2026
(Anthropic · Cerebral Valley · Gladstone Institutes; 1 of 299)
GitHub | Demo Video

- Calibrated MCP triage tool for non-coding cancer variants, integrating chromatin (ChromBPNet), evolutionary constraint and MPRA function converged with base-level mechanism; returns a confidence flag on every call.

**Unravel - 1st place, Fivetran track, Google Cloud Rapid Agent Hackathon** Jul 2026
GitHub | Live Link | Demo Video

- Five-agent Gemini 3.1 / Google ADK system that re-scores ACMG pathogenicity when a variant is reclassified and drafts clinician-approved patient recontact (FHIR).

**RegulonForge - Winner (1 of 50 winners from 4,097 submissions)** Apr 2026
(Google DeepMind - Vibe Code with Gemini 3 Pro in AI Studio)
Live Link | Demo Video

- Multimodal regulatory-genomics simulator (Gemini 3 Pro) pairing sequence analysis with genome-browser image interpretation; includes a perturbation sandbox and a teaching mode with interactive quizzes.

## CONFERENCES, PRESENTATIONS & TRAINING

- **YC Startup School Paris, Y Combinator (Jun 2026) -** ~700 selected from 4,000+ applications.
- **Harvard Health Systems Innovation Lab (HSIL) Hackathon (Apr 2026)** - built CareSignal, an ambient AI care scribe for domiciliary care (CQC-compliant notes, FHIR/SNOMED).
- **NVIDIA Deep Learning Institute Workshop (Nov 2025)** - Introduction to Deep Learning, Autoencoders and PyTorch (CECAM-IRL, University College Dublin).
- **CECAM Flagship School: Mastering Dimensionality Reduction with Autoencoders (May 2026)** - competitive advanced tutorial on autoencoders, variational autoencoders (VAEs) and graph neural networks in PyTorch (CECAM-IRL / ICHEC, University College Dublin).
- **Imperial College London Robotics Summer School (Jul 2026) -** Hamlyn Centre for Robotic Surgery; led model interpretability (HiResCAM, with causal and sanity checks) on a team pipeline that keeps a confocal endomicroscopy probe auto-focused on tissue for real-time tumour-margin imaging.
- **Conference Posters:** All-Ireland Chromatin Consortium (2026); Conway Institute Research Festival, UCD (2025); RCSI Research Day (2026); BRS Research Day, DCU (2025).

## ACADEMIC DISTINCTIONS & AWARDS

- **Class ranking, BSc Genetics & Cell Biology (DCU):** ranked 1st in class in Years 1-2 and 2nd in class in Years 3-4; graduated 2nd overall (Salutatorian), Dean's Honour List.
- **Undergraduate Academic Achievement Award, DCU:** Awarded to highest performing second year students across DCU departments.
- **FSH Academic Scholarship, DCU:** awarded on a CAO entry score of 550+ points.
- **Poster Commendation, UCD Conway Institute Research Festival (Oct 2025):** scored a perfect 30/30.

## PROFESSIONAL MEMBERSHIP

Member, European Association for Cancer Research (EACR).
Member, Irish Association for Cancer Research (IACR).

## MENTORING, OUTREACH & SCIENCE COMMUNICATION

- **Undergraduate research projects, RCSI (2027) -** designed three six-week computational projects for third-year medical students on somatic mutation and the breast chromatin atlas, two of which are offered in the January and February 2027 blocks.
- **Alumni Mentor, DCU Career Mentoring Programme (October 2026 to March 2027):** one-to-one career mentoring of a second-year undergraduate.
- **Green Computing Representative, RCSI Green Labs Committee (Sep 2026–present)** - nominated to the departmental sustainability committee to assess and reduce the carbon footprint of research computing (HPC, cloud, and model training).
- **Invited speaker, Anthropic Claude Science AMA (Aug 2026) -** live demo and Q&A on deploying AI agents in cancer epigenomics research, alongside Claude Science lead engineer, Alexander Tarashansky.
- **Invited panel speaker, AMD-sponsored UCD STEM careers panel (Apr 2026) -** STEM careers, academia versus industry, and the future of computational biology.
- **Exhibition Contributor, DCU (Oct 2025) -** selected for a university showcase recognising outstanding student research and leadership.
- **Best Lay Presentation Award, Breakthrough Cancer Research (Aug 2025)** - for communicating complex research to a lay panel of cancer survivors and research peers.
- **Student Blog Contributor, DCU & Unibuddy (Feb - Apr 2025)** - authored guidance for prospective Genetics & Cell Biology students.

## SKILLS

**Programming:** Python, R, Bash, Perl, TypeScript

**Machine Learning & AI:** PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, deep learning (CNNs, MLPs), multi-agent systems (CrewAI, Google ADK), RAG, LangChain

**Genomics & Data Science:** NGS, single-cell ATAC-seq, cfDNA/methylation & bisulfite analysis, survival analysis (Kaplan–Meier, Cox), NMF/NNLS deconvolution, TCGA/DepMap/ENCODE, Docker

**Cloud & App:** Google Cloud Platform, Vertex AI, FastAPI, Streamlit, React.js, Firebase

**Languages:** English (Native), French (Professional Working Proficiency), Irish (Intermediate Reading, Basic Spoken), Yoruba (Receptive Bilingualism)